

EX.NO : 03

DATE :

**ANALYZING THE PERFORMANCE OF VARIOUS
CONFIGURATIONS AND PROTOCOLS IN LAN CONNECTING TWO
LANs USING ROUTER WITH STATIC ROUTER**

AIM :

To establish connection between two LANs by extending routing connection using router.

REQUIREMENTS :

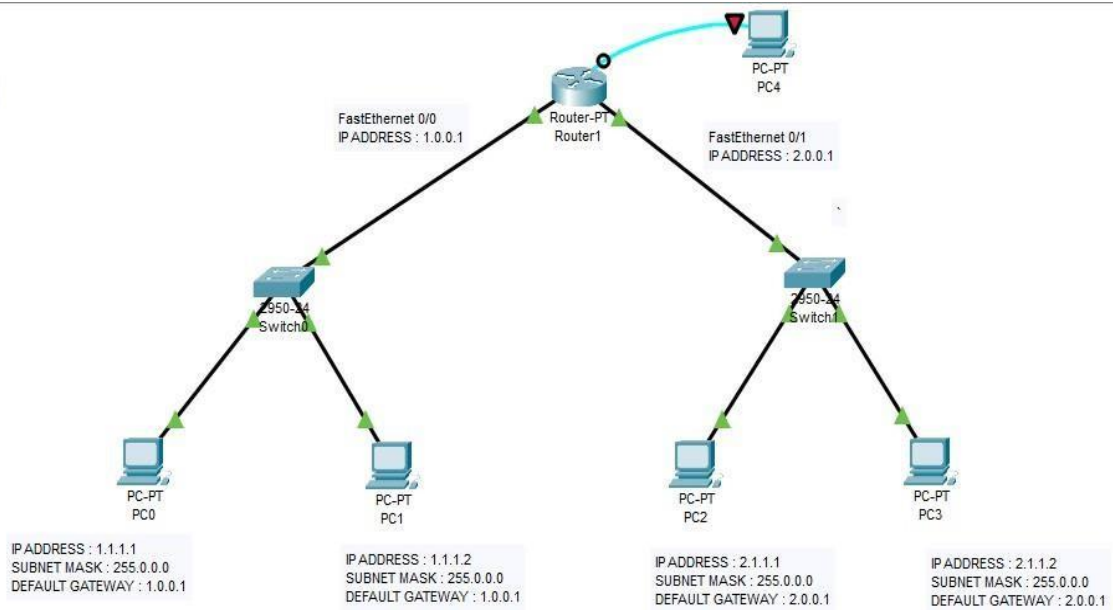
- Five windows PC.
- Two Switch (2950-24).
- Six Straight Line LAN Cables.
- One Router (Router PT).
- One console connection of router with PC to configure router.
- Cisco Packet Tracer.

PROCEDURES :

- Open CISCO PACKET TRACER software.
- Draw the Five PC using END Device Icons.
- Draw the Cisco 24 Port Switch Using Switch icon lists.
- Draw the Cisco Generic Routers using Router icon lists.
- Make the Connections using Copper-Straight-Through Ethernet Cables.
- Enter the IP Address To Each Machine.
- Configure Router PT – 0.
- Check the Network Connections using Add Simple PDU(P).

NETWORK TOPOLOGY :

EX_NO : 03



HOST PC IP ADDRESS :

PC0 IP CONFIGURATION

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 1.1.1.1

Subnet Mask 255.0.0.0

Default Gateway 1.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

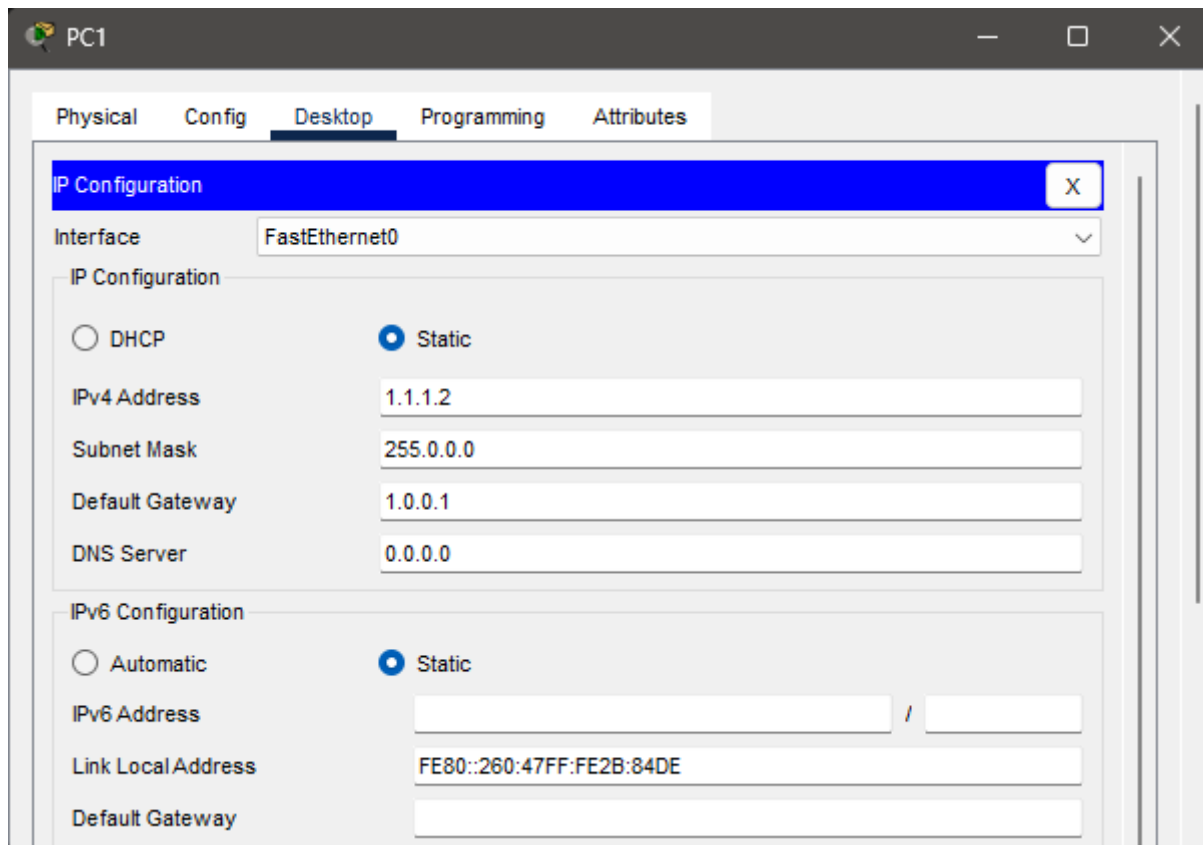
☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::290:2BFF:FE77:CC8

Default Gateway

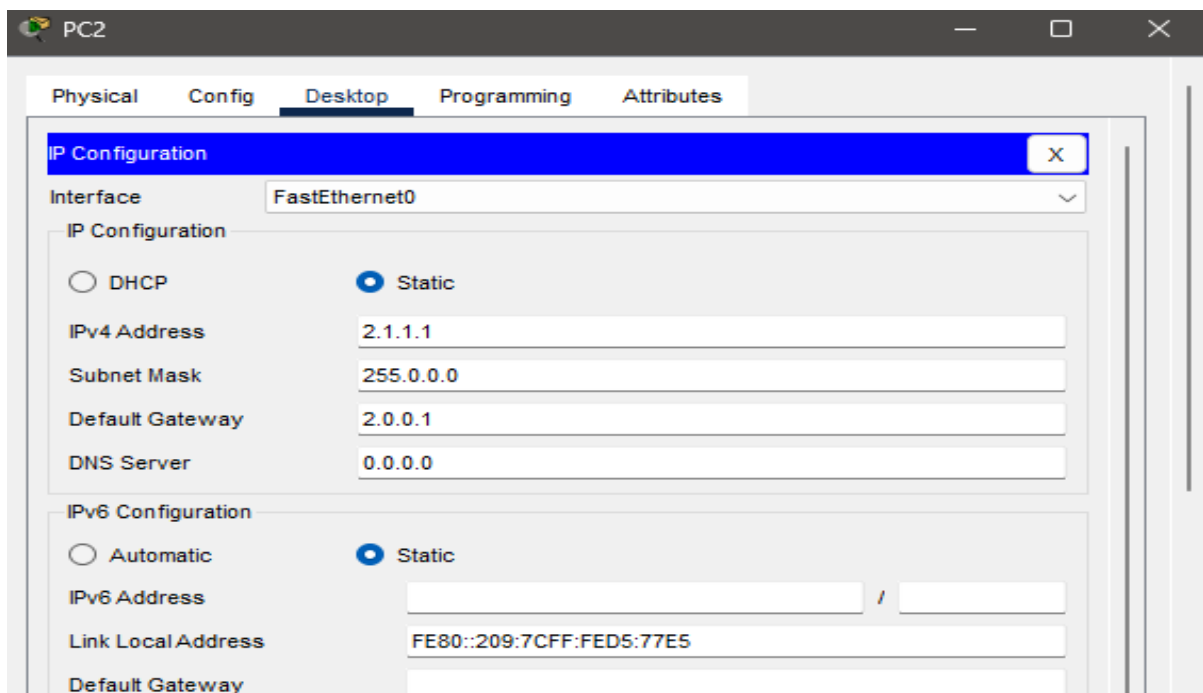
PC1 IP CONFIGURATION :



The screenshot shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 1.1.1.2, Subnet Mask: 255.0.0.0, Default Gateway: 1.0.0.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The fields are filled with: IPv6 Address: (empty), Link Local Address: FE80::260:47FF:FE2B:84DE, and Default Gateway: (empty).

Configuration Type	Field	Value
IP Configuration (Static)	Interface	FastEthernet0
	IPv4 Address	1.1.1.2
	Subnet Mask	255.0.0.0
	Default Gateway	1.0.0.1
IPv6 Configuration (Static)	IPv6 Address	
	Link Local Address	FE80::260:47FF:FE2B:84DE
	Default Gateway	
	DNS Server	0.0.0.0

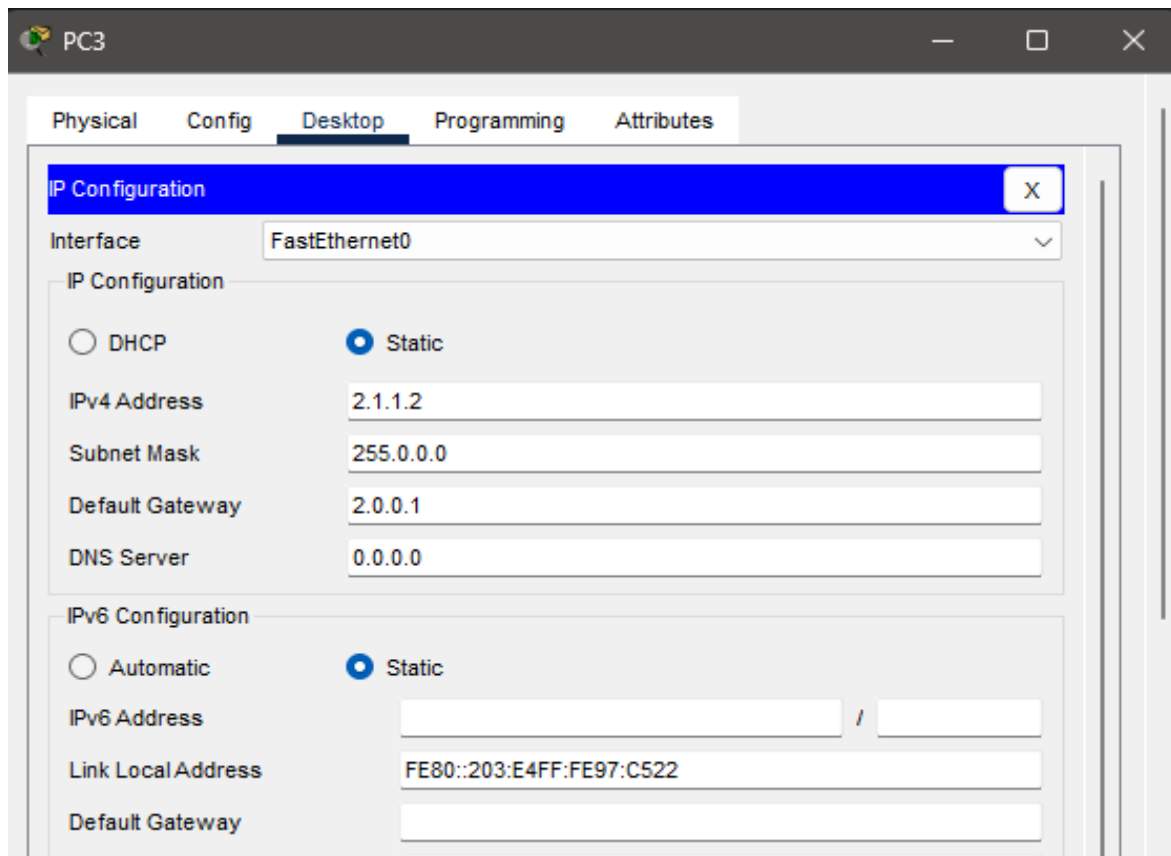
PC2 IP CONFIGURATION :



The screenshot shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 2.1.1.1, Subnet Mask: 255.0.0.0, Default Gateway: 2.0.0.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The fields are filled with: IPv6 Address: (empty), Link Local Address: FE80::209:7CFF:FED5:77E5, and Default Gateway: (empty).

Configuration Type	Field	Value
IP Configuration (Static)	Interface	FastEthernet0
	IPv4 Address	2.1.1.1
	Subnet Mask	255.0.0.0
	Default Gateway	2.0.0.1
IPv6 Configuration (Static)	IPv6 Address	
	Link Local Address	FE80::209:7CFF:FED5:77E5
	Default Gateway	
	DNS Server	0.0.0.0

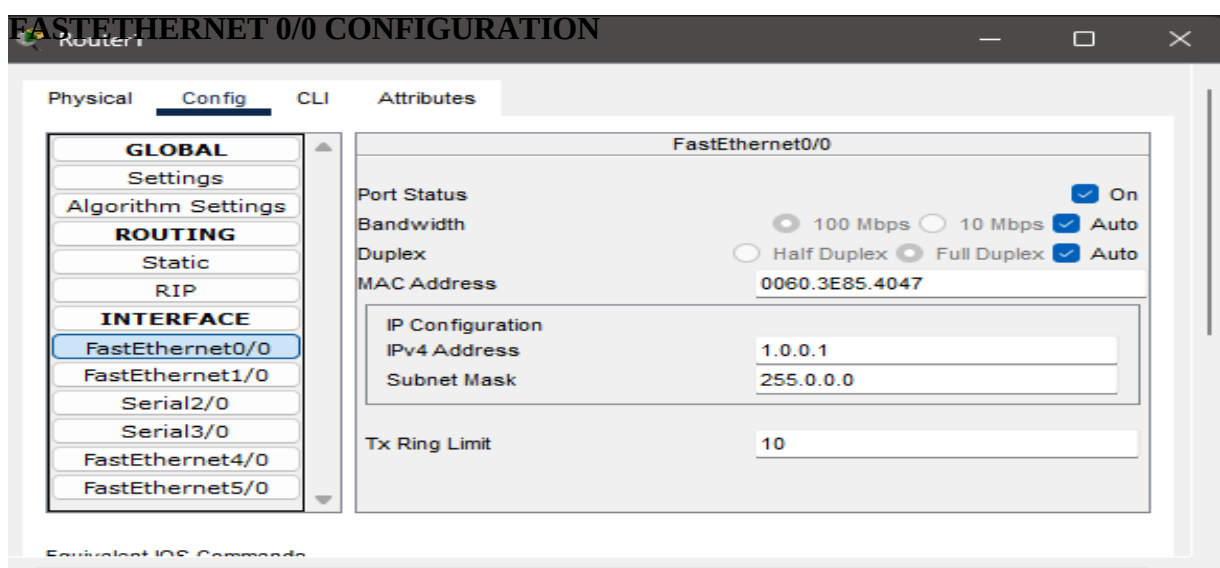
PC3 IP CONFIGURATION :



The image shows a window titled "PC3" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, and the "IP Configuration" window is open. It shows the configuration for the "FastEthernet0" interface. The "Static" radio button is selected under "IP Configuration". The "IPv4 Address" is set to "2.1.1.2", "Subnet Mask" is "255.0.0.0", "Default Gateway" is "2.0.0.1", and "DNS Server" is "0.0.0.0". Under "IPv6 Configuration", the "Static" radio button is also selected. The "IPv6 Address" field is empty, "Link Local Address" is "FE80::203:E4FF:FE97:C522", and "Default Gateway" is empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	2.1.1.2
Subnet Mask	255.0.0.0
Default Gateway	2.0.0.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::203:E4FF:FE97:C522
Default Gateway	

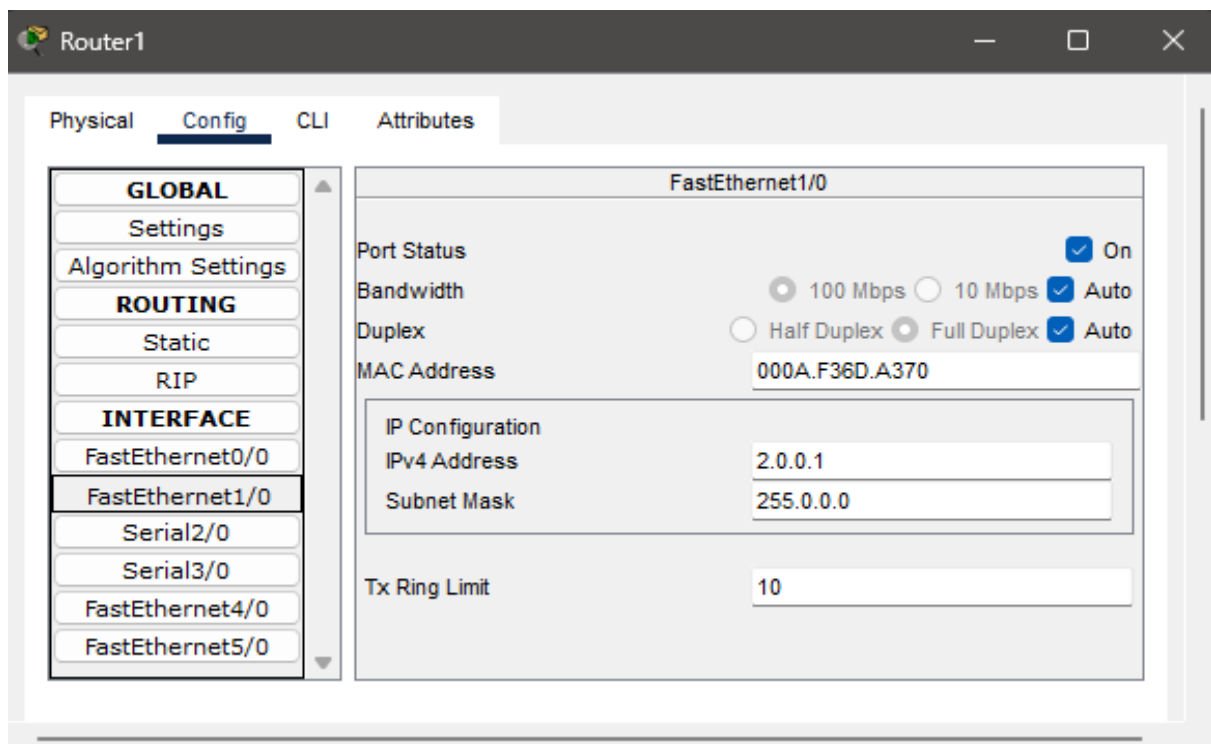
ROUTER 0 CONFIGURATION :



The image shows a window titled "Router 0" with tabs for Physical, Config, CLI, and Attributes. The "Config" tab is active, and the "FastEthernet0/0" configuration window is open. The "FastEthernet0/0" interface is selected in the left sidebar. The "Port Status" is "On", "Bandwidth" is "100 Mbps", "Duplex" is "Full Duplex", and "MAC Address" is "0060.3E85.4047". The "IP Configuration" section shows "IPv4 Address" as "1.0.0.1" and "Subnet Mask" as "255.0.0.0". The "Tx Ring Limit" is set to "10".

FastEthernet0/0	
Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0060.3E85.4047
IP Configuration	
IPv4 Address	1.0.0.1
Subnet Mask	255.0.0.0
Tx Ring Limit	10

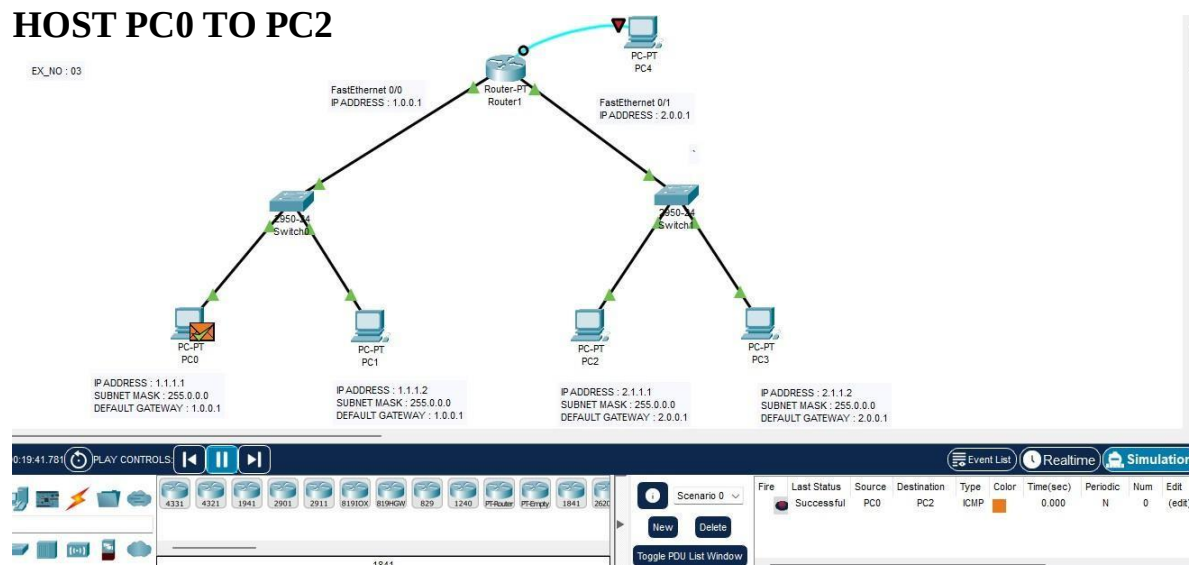
FASTETHERNET 1/0 CONFIGURATION



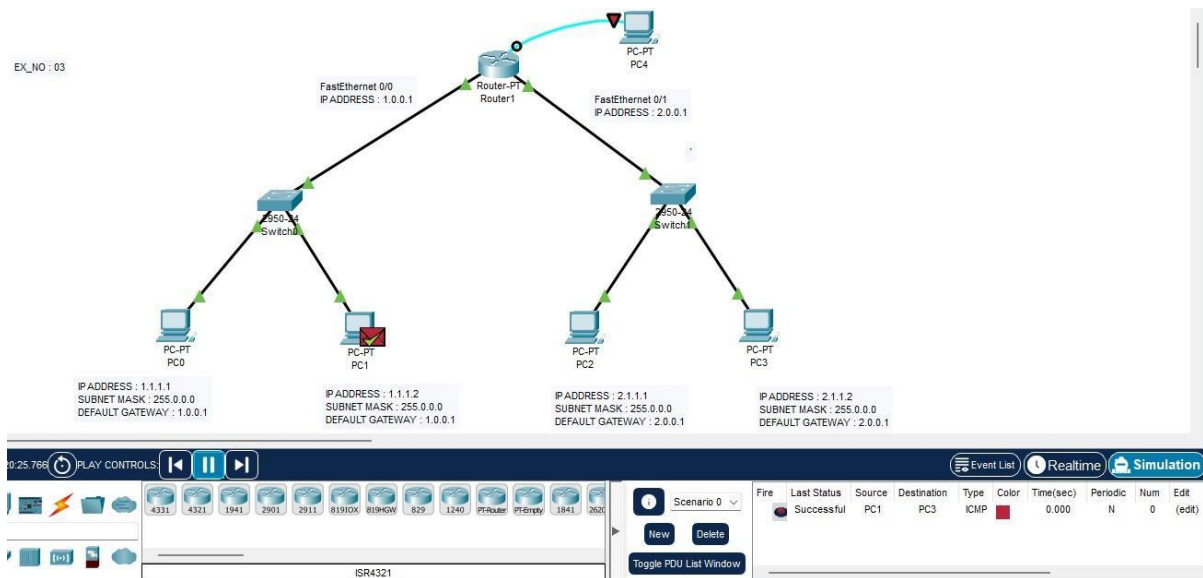
VERIFY LAN NETWORK CONNECTIVITY :

Using Add Simple PDU(p), Click the mail icon and then drop one mail to one of the PC in first lan and another mail to PC in another lan. If the resultant window shows the successful delivery, then network connectivity is successful.

HOST PC0 TO PC2



HOST PC1 TO PC3



RESULT :

Thus, two LANs are connected using router with static router and the communication between LANs is checked successfully.